

Abstract:

RailSmart project aims to revolutionize the traditional railway ticketing system by developing an automated platform for ticket identification, validation, and violation detection. The system integrates advanced machine learning models, such as TensorFlow and PyTorch, to efficiently detect ticket violations and fraud. Utilizing technologies like Tesseract OCR for ticket scanning and Flutter or React for user registration, RailSmart ensures that each passenger's ticket is securely linked to their identity, reducing instances of unauthorized travel. The platform also incorporates GPS-based validation and geo-fencing for real-time monitoring, enhancing the security and accuracy of the ticketing process. Moreover, RailSmart is designed to streamline operations through seamless integration with widely-used payment gateways such as Paytm and Google Pay, ensuring fast, transparent, and secure payment processing for both ticket purchases and fine collection. The system supports automated e-challan issuance for violations, utilizing CCTV and AI-based monitoring systems to enforce compliance and improve operational efficiency. By offering a comprehensive dashboard with real-time analytics on ticketing trends, passenger behaviour, and violation patterns, RailSmart enables railway authorities to make data-driven decisions that enhance revenue management and passenger experience. Ultimately, RailSmart provides a modern, scalable, and secure solution that addresses the inefficiencies of current manual and digital railway ticketing systems, offering significant improvements in terms of fraud prevention, operational transparency, and passenger convenience.